

ZANDER STRAIT

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EDUCATION

Stony Brook University

Bachelor of Science, Computer Science

Stony Brook, NY

Expected May 2027

Courses: Data Structures & Algorithms, Fundamentals of Software Development, Analysis of Algorithms, Computational Geometry, Mathematical Logic, Programming Abstractions, Systems-Level Programming

PROJECTS

Media Scout | *Next.js, TypeScript, Tailwind CSS*

Nov 2025 – Present

[Source Code](#) | [Live Demo](#)

- Developed a full-stack marketplace aggregation platform, enabling users to instantly find and compare physical media listings across multiple vendors such as Discogs and eBay.
- Built a backend aggregation engine that concurrently fetches and paginates results from multiple third-party **RESTful APIs**, normalizing their inconsistent data structures into a single unified schema.
- Integrated an API caching layer that reduced repeated search latency by **99% (2.5s to < 20ms)** while minimizing unnecessary third-party API calls.
- Implemented dynamic filtering and sorting with React's **useMemo** to efficiently process combined datasets client-side, eliminating redundant network requests.
- Leveraged the Last.fm API to power real-time album search autocomplete with cover art previews.
- Created a persistent watchlist using custom React hooks and **localStorage**, retaining a user's favorited listings across sessions without requiring backend authentication.

StereoFM | *React, Javascript, MongoDB, Express, Node*

Sep 2025 – Jan 2026

[Source Code](#) | [Live Demo](#)

- Engineered a full-stack social music platform, including a robust RESTful API designed to manage user profiles, custom playlists, and a scalable catalog of **5,000+** tracks.
- Implemented secure, stateless authentication using **JWT** and **bcrypt**, enforcing granular data ownership permissions to ensure users can only modify their own contributed content.
- Designed a dynamic frontend with **Material UI**, featuring server-side filtering, sorting, and integrated YouTube IFrame API playback with automated error handling.
- Developed an auto-reset demo environment that reseeds data on login to enable testing without registration.

Live LaTeX Editor | *React, Javascript, CSS*

May 2025 – Nov 2025

[Source Code](#) | [Live Demo](#)

- Developed a single-page LaTeX editor utilizing the **KaTeX** engine for instantaneous client-side compilation, rendering at speeds that significantly outperform alternatives like MathJax.
- Architected a modular component system with **unidirectional data flow**, managing cross-component state to synchronize font styling, alignment, and formatting configurations.
- Engineered a non-blocking rendering pipeline using **input debouncing**, optimizing application performance to prevent main-thread blocking and ensure UI responsiveness during rapid typing.
- Integrated and customized the **CodeMirror** library to construct a robust code editor, implementing custom syntax highlighting and complex logic to handle dynamic text insertions from external UI controls.

EXPERIENCE

Operations Associate

May 2023 – Jul 2023

Kunz Greenhouses

Port Jefferson, NY

- Collaborated in a fast-paced environment to deliver customers tailored solutions and recommendations.

SKILLS

Professional Skills: Web Development, Graphic Design, Git/GitHub, API Development, UI/UX, Figma

Languages / Frameworks: React, Next.js, TypeScript, JavaScript, Python, Java, Express, MongoDB, PostgreSQL, LaTeX, Tailwind, HTML/CSS

Interests / Hobbies: Art, Music, Photography, Video Editing, Film, Reading, Guitar